TEACHER GUIDE · KINDERGARTEN · SEQUENCING & ALGORITHMS

Follow the Arrows

Quick facilitation notes & answer key

Goal

Children execute a sequence of ➡ moves on a 1D board and report Bit's landing box. Carrying out step-by-step directions in order is executing an algorithm — and the count of steps determines the result.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children follow a sequence of directional instructions.

No reading required of the child: read each prompt aloud; the child writes the landing box number.

Materials

The printed worksheet + a pencil. Optional: floor boxes 1-5 with a star on box 4 to walk the steps.

Run it (about 15 min)

1. MODEL: from box 1, walk ➡ ➡ ➡ aloud — box 2, box 3, box 4. Bit lands on the star.
2. GUIDED: Exercise 1 — two steps from box 1 lands on box 3.
3. YOU TRY: children do four steps (Ex 2 → box 5), then start from box 2 (Ex 3 → box 4, the star).
4. Connect: 'Computers follow their steps in order, exactly — just like Bit.'

Answer key (modelled by the run-gate)

Model — start 1, ➡ ➡ ➡ → box 4 (the star).

Ex 1 — start 1, ➡ ➡ → box 3.

Ex 2 — start 1, ➡ ➡ ➡ ➡ → box 5.

Ex 3 — start 2, ➡ ➡ → box 4 (yes, the star).

Watch for & differentiate

Common slip: counting box 1 as the first move — the first ➡ moves Bit OFF box 1 to box 2.

Simplify: walk each step on the floor and stop to name the box.

Extend: ask how many steps from box 1 reach box 5 (4), or start from box 3.

Movement option: call out arrows and have children take one step each, freezing on the landing box.

Success looks like

The child counts the ➡ steps from the start box and writes the correct landing box for each program.